1 **(a)** $360 \div 24$ **[M1]**

15 **[A1]**

(b) $(15 - 2) \times 180$ **[M1]**

2340 **[A1]**

(c) Exterior angle $= 180 - 165 = 15°$ **[M1]**

$360 \div 15 = 24$ **[A1]**

[Total Marks: 6]

2 **(a)** 125 **[B1]**

(b) Corresponding angles (are equal) **[B1]**

(c) 144 **[B1]**

[Total Marks: 3]

3 **(a)** $180 - (55 + 70)$ **[M1]**

55 **[A1]**

(b) 55 **[B1]**

(c) Angle $DAE = 55°$ (from part a) AND Angle $ADE = 55°$ **[B1]**

Two angles are equal, so the triangle is isosceles **[B1]**

[Total Marks: 5]

4 **(a)** $5x + 10 = 7x - 30$ **[M1]**

$x = 20$ **[A1]**

Corresponding angles (are equal) **[B1]**

(b) Angle $GHD = 7(20) - 30 = 110°$ **[B1]**

Angle $GHK = 180 - 110 = 70°$ (Angles on a straight line) **[B1]**

Angle $GKH = 180 - (70 + 35) = 75°$ (Angles in a triangle) **[B1]**

[Total Marks: 6]

5 **(a)** $(5 - 2) \times 180 \div 5$ OR $180 - (360 \div 5)$ **[M1]**

108 **[A1]**

(b) Interior angle of square $= 90°$ **[B1]**

Interior angle of n-gon $= 360 - 90 - 108 = 162°$ **[M1]**

$n = 360 \div (180 - 162) = 20$ **[A1]**

(c) Recognises triangle PFA is isosceles (since $PA = PF$ from shared sides of regular polygons) **[M1]**

Angle $PFA = (180 - 162) \div 2 = 9$ **[A1]**

[Total Marks: 7]

6 **(a)** $\dfrac{(5-2) \times 180}{5}$ OR $180 - \dfrac{360}{5}$ **[M1]**

108 **[A1]**

(b) Interior angle of hexagon $= 120°$ **[B1]**

Angle $AEI = 360 - 108 - 120 = 132$ **[A1]**

(c) $AE = EI$ (sides of regular polygons sharing an edge are equal) **[M1]**

Triangle AEI is isosceles **[M1]**

Angle $EAI = \dfrac{180 - 132}{2} = 24$ **[A1]**

[Total Marks: 7]